

Consider the linear time-invariant system

$$x(k+1) = \underbrace{\begin{bmatrix} 1/2 & -1/4 \\ 1/4 & 1/2 \end{bmatrix}}_A x(k) + \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_B u(k)$$

subject to state constraints $\|x(k)\|_\infty \leq 1$ and input constraints $|u(k)| \leq 1$. The MPC formulation for the general reference tracking case with $Q = I_2$ and $R = 1$ is given as follows.

$$\begin{aligned} \underset{U}{\text{minimize}} \quad & (x_N - x_s)^\top P (x_N - x_s) + \sum_{i=0}^{N-1} (x_i - x_s)^\top (x_i - x_s) + (u_i - u_s)^\top (u_i - u_s) \\ \text{subject to} \quad & x_{i+1} = Ax_i + Bu_i, \\ & x_i \in \mathcal{X} = \{x \mid \|x\|_\infty \leq 1\}, \\ & u_i \in \mathcal{U} = \{u \mid |u| \leq 1\}, \\ & x_N \in \{x_s\} \oplus \mathcal{X}_f, \\ & x_0 = x(k). \end{aligned} \tag{1}$$

Exercise 1 (Regulation Case)

Consider first the regulation case, i.e., $x_s = 0$ and $u_s = 0$, and a zero terminal control law $\kappa_f(x) = 0$. Show that the choice of

- terminal set $\mathcal{X}_f = \{x \mid x^\top x \leq 1\}$, and
- terminal cost weighting $P = aI$, with scalar parameter $a > 0$, that solves the Lyapunov equation $A^\top P A - P = -I$,

results in a recursively feasible and asymptotically stabilizing MPC controller.

Solution:

Assumptions (Lec. 6, Slide 42):

1. $Q = I$ and $R = 1$ are positive definite $\Rightarrow$ Stage cost positive definite $\checkmark$
 2. (a) $\{x \mid x^\top x \leq 1\}$ is a level set of the Lyapunov function $V(x) = x^\top P x \Rightarrow \mathcal{X}_f$ invariant $\checkmark$
(b) $\mathcal{X}_f \subseteq \mathcal{X}$ and $\kappa_f(x) = 0 \in \mathcal{U} \Rightarrow$ Terminal constraint satisfaction $\checkmark$
 3. $x^\top A^\top P A x - x^\top P x \leq -x^\top x - \kappa_f(x)^\top \kappa_f(x) \Rightarrow$ Terminal cost decrease $\checkmark$
- $\Rightarrow$ Recursively feasible and asymptotically stable.

Exercise 2 (Reference Tracking)

Consider the reference tracking problem with the system output $y(k) = Hx(k) \rightarrow r$ with $H = [1 \ 0]$ and reference $r = 0.25$.

- i) Compute the resulting steady state x_s and corresponding input u_s .
- ii) Modify the terminal set $\mathcal{X}_f$ such that the MPC problem (1) is recursively feasible and accurate tracking is guaranteed.

Solution:

i) x_s and u_s need to satisfy

- steady state: $x_s = Ax_s + Bu_s$
- reference tracking $r = Hx_s$

$$\begin{bmatrix} I - A & -B \\ H & 0 \end{bmatrix} \begin{bmatrix} x_s \\ u_s \end{bmatrix} \stackrel{!}{=} \begin{bmatrix} 0 \\ r \end{bmatrix} \Leftrightarrow \begin{bmatrix} 1/2 & 1/4 & 0 \\ -1/4 & 1/2 & -1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_{s,1} \\ x_{s,2} \\ u_s \end{bmatrix} \stackrel{!}{=} \begin{bmatrix} 0 \\ 0 \\ 1/4 \end{bmatrix}$$

- $x_{s,1} = r = 1/4$
 - $1/2 \cdot x_{s,1} + 1/4 \cdot x_{s,2} \stackrel{!}{=} 0 \rightarrow x_{s,2} = -1/2$
 - $-1/4 \cdot x_{s,1} + 1/2 \cdot x_{s,2} - u_s \stackrel{!}{=} 0 \rightarrow u_s = -1/16 - 1/4 = -5/16$
- $\Rightarrow x_s = [1/4 \ -1/2]^\top$ and $u_s = -5/16$

ii) Sketch state constraints with steady state and terminal set.

- resulting tracking MPC terminal constraint $x_N \in \{x_s\} \oplus \mathcal{X}_f \not\subseteq \mathcal{X}$
 - however, a scaled version of $\mathcal{X}_f$ maintains all the necessary properties (invariance and cost decrease, see Lec 7 Slide 22)
- $\Rightarrow$ rescale such that $\{x_s\} \oplus \alpha \mathcal{X}_f \subseteq \mathcal{X}$: By inspection $\alpha = 0.5$.

Exercise 3 (Disturbance Estimation)

Consider now that the system is subject to time-invariant disturbances d

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k) + d \\ y(k) &= x(k) \end{aligned}$$

where we assume to be able to measure the full state $x(k)$.

- Derive the estimation-error dynamics for the augmented state $\tilde{x} = [x(k) \ d]^\top$ under a linear estimator L
- Design a stable estimator gain L in Matlab (e.g., using pole placement)
- Simulate the autonomous system from $x(0) = \hat{x}(0) = [1 \ 1]^\top$ with $d = [0.1 \ -0.2]^\top$ and $\hat{d}(0) = [0 \ 0]^\top$.

Solution:

- Augmented system dynamics:

$$\begin{aligned} \begin{bmatrix} x(k+1) \\ d \end{bmatrix} &= \underbrace{\begin{bmatrix} A & I \\ 0 & I \end{bmatrix}}_{A_{\text{aug}}} \begin{bmatrix} x(k) \\ d \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} u(k) \\ y(k) &= \underbrace{\begin{bmatrix} I & 0 \end{bmatrix}}_{C_{\text{aug}}} \begin{bmatrix} x(k) \\ d \end{bmatrix} \end{aligned}$$

- Linear observer to estimate the state $\hat{x}(k)$ and disturbance $\hat{d}(k)$

$$\begin{aligned} \begin{bmatrix} \hat{x}(k+1) \\ \hat{d}(k+1) \end{bmatrix} &= \begin{bmatrix} A & I \\ 0 & I \end{bmatrix} \begin{bmatrix} \hat{x}(k) \\ \hat{d}(k) \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} u(k) - \begin{bmatrix} L_x \\ L_d \end{bmatrix} (y(k) - \hat{y}(k)) \\ \begin{bmatrix} \hat{x}(k+1) \\ \hat{d}(k+1) \end{bmatrix} &= \begin{bmatrix} A & I \\ 0 & I \end{bmatrix} \begin{bmatrix} \hat{x}(k) \\ \hat{d}(k) \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} u(k) - \begin{bmatrix} L_x \\ L_d \end{bmatrix} [I \ 0] \left(\begin{bmatrix} x(k) \\ d \end{bmatrix} - \begin{bmatrix} \hat{x}(k) \\ \hat{d}(k) \end{bmatrix} \right) \end{aligned}$$

- Introduce error dynamics by subtracting observer dynamics from state dynamics:

$$\begin{aligned} \begin{bmatrix} x(k+1) - \hat{x}(k+1) \\ d - \hat{d}(k+1) \end{bmatrix} &= \begin{bmatrix} A & I \\ 0 & I \end{bmatrix} \begin{bmatrix} x(k) - \hat{x}(k) \\ d - \hat{d}(k) \end{bmatrix} + \begin{bmatrix} L_x \\ L_d \end{bmatrix} \begin{bmatrix} I & 0 \end{bmatrix} \begin{bmatrix} x(k) - \hat{x}(k) \\ d - \hat{d}(k) \end{bmatrix} \\ &= \underbrace{\left(\begin{bmatrix} A & I \\ 0 & I \end{bmatrix} + \begin{bmatrix} L_x \\ L_d \end{bmatrix} \begin{bmatrix} I & 0 \end{bmatrix} \right)}_{A_e} \begin{bmatrix} x(k) - \hat{x}(k) \\ d - \hat{d}(k) \end{bmatrix} \end{aligned}$$

- ii) • Error dynamics asymptotically stable if A_e has eigenvalues contained within the unit circle.
- Find $[L_x \ L_d]^\top$, e.g., using pole placement (see Matlab script)
- iii) See Matlab script